

STEADY

Personalized Recovery Trajectory

Implementation-Ready Engineering Handoff

Clinical Intelligence Expansion | Document 04 | Repository-grounded against tprawlins-lang/Emdr | branch
claude/gifted-keller-501y5d | September 2026

Build directive

Extend Steady from descriptive longitudinal charts into patient-specific recovery-course intelligence. The system identifies meaningful slowing, stalling, or reversal within separate clinical/function domains, explains the evidence, and emits non-safety review signals to the Care Command Center. It is not a predictive risk score and it does not promise an expected outcome.

Shared product rule

The clinician decides what clinical meaning and action to accept. Steady may measure, organize, compare, and suggest, but it does not silently turn an association, model output, or behavioral signal into clinical fact.

Repository foundation this handoff assumes

Foundation	Current Steady anchor	Rule for this build
Longitudinal history	src/lib/spine.ts + ADR 0010	Durable state must be event-backed and replayable. Do not create a parallel mutable history.
Tenant/person boundary	src/lib/repository.ts + ADR 0011	Every new durable patient record carries tenant_id and person_id; new paths use TenantContext even while legacy paths are migrating.
Clinician patient workspace	src/app/clinician/member/[id]/*	New clinician surfaces extend PersonShell and existing evidence/review conventions.
Evidence-linked summary	src/lib/clinical/summary.ts	A generated claim without resolvable evidence is withheld. Summary is a reading aid, not the record.
Trajectory	src/lib/clinical/trajectory.ts	Separate scales stay separate; provenance remains visible; do not invent a composite clinical meaning from arbitrary axis alignment.
Work queue	src/lib/clinical/work-queue.ts	Clinician home opens on actionable work, stable server ordering, clear reason, one primary action.
Companion memory	src/lib/companion.ts	Patient/Companion data retains source class and memory exposure rules. AI output is not clinician evidence.
AI Gateway	ADR 0012	New model tasks must route through the gateway with schema validation, task version, provenance, evaluation, and PHI policy.

This series also assumes the Clinician Thoughts / Clinical Memory / Session Prep Engineering Specification v2.1 is implemented or being implemented as the common clinician-memory and evidence layer.

Cross-feature invariants

- **Safety authority stays deterministic.** These features may add clinician decision support but cannot clear, weaken, bypass, or replace the safety engine.
- **Stabilization scope stays intact.** No feature autonomously initiates trauma reprocessing, changes a gated treatment decision, diagnoses, or issues a treatment order.
- **Source classes never collapse.** Patient report, clinician observation, system measurement, formal note, Companion interaction, and model inference remain distinguishable.
- **Associations are associations.** A temporal or statistical relationship is not described as causation unless a clinician explicitly records that judgement from appropriate evidence.
- **No future-data leakage.** Historical reconstruction and point-in-time evaluations use only information Steady knew at the relevant cutoff.
- **Missing data stays visible.** Missing observations do not become zero, normal, compliant, or recovered.
- **Corrections append.** An authorized correction supersedes prior derived state without erasing history.
- **Model output is disposable; evidence and provenance are permanent.** Every material intelligence output points to evidence IDs and task/policy versions.

1. Why this is an extension, not a rewrite

- The existing trajectory module already correctly keeps activation, sleep, PHQ-9, GAD-7, PCL-5 and other measures on separate scales and preserves reconstructed-history provenance.
- Do not replace that chart with one composite "recovery score." Add a domain-analysis layer above it.
- Return-to-Life becomes the function domain. Response Fingerprint supplies recovery-burden and intervention-response context. Existing measures/check-ins remain their own domains.

2. Trajectory domains

Domain	Primary evidence	Direction semantics
function	Return-to-Life accepted observations	Goal-specific movement; no averaging unrelated goals into one clinical score by default.
activation	daily check-in activation	Lower may be better only in context; retain existing measure meaning.
dissociation	daily check-in dissociation	Lower generally indicates less reported dissociation.
sleep	sleep-quality check-ins	Higher quality is better on existing scale.
validated measures	PHQ-9, GAD-7, PCL-5, ITQ etc.	Each instrument stays on its own validated scale.
session recovery	session response + delayed recovery burden	Mixed immediate/delayed pattern allowed.
engagement	presence/gaps	Descriptive context only; not adherence or prediction.

3. State model

State	Meaning	Display requirement
insufficient_data	Not enough comparable observations	Say what is missing.
improving	Meaningful favorable movement across the defined comparison window	Show start/end evidence and window.
stable	No meaningful movement beyond noise threshold	Do not call "not improving."
slowing	Previously favorable movement has materially reduced across repeated windows	Describe as trajectory change, not failure.
stalled	Meaningful target/domain has remained within a narrow band across the configured review window	Requires adequate observation density.
reversing	Recent repeated movement is opposite the prior favorable direction	Requires more than one adverse point unless policy says otherwise.

4. Analysis policy v1

- Use deterministic, explainable rules first. Do not start with an opaque predictive model.
- Minimum density and thresholds are versioned per domain. Example defaults can be seeded for demo but must be configuration, not hidden constants.
- Use robust comparison windows, such as median of recent observations versus prior window, rather than a single point-to-point delta for deviation states.
- Require persistence for slowing/stalled/reversing. One bad day remains one observation.
- Do not infer causation from co-occurring thread/intervention data. Context can be displayed as "occurred during" or "also changed."
- A patient can improve in one domain and worsen in another. Preserve the disagreement.

5. Database schema

```
CREATE TABLE recovery_trajectory_snapshots (
  id TEXT PRIMARY KEY,
  tenant_id TEXT NOT NULL,
  person_id TEXT NOT NULL,
  domain_type TEXT NOT NULL,
  domain_key TEXT NOT NULL,
  state TEXT NOT NULL CHECK (state IN (
    'insufficient_data', 'improving', 'stable', 'slowing', 'stalled', 'reversing'
  )),
  policy_version TEXT NOT NULL,
  evidence_cutoff TEXT NOT NULL,
  current_window_json TEXT NOT NULL,
  comparison_window_json TEXT,
  explanation_json TEXT NOT NULL,
  limitations_json TEXT NOT NULL DEFAULT '[]',
```

```

    computed_at TEXT NOT NULL,
    UNIQUE(tenant_id, person_id, domain_type, domain_key, policy_version, evidence_cutoff)
);

CREATE TABLE recovery_trajectory_evidence (
    snapshot_id TEXT NOT NULL,
    evidence_type TEXT NOT NULL,
    evidence_id TEXT NOT NULL,
    rank INTEGER NOT NULL,
    PRIMARY KEY(snapshot_id, evidence_type, evidence_id)
);

CREATE TABLE recovery_trajectory_reviews (
    id TEXT PRIMARY KEY,
    tenant_id TEXT NOT NULL,
    person_id TEXT NOT NULL,
    snapshot_id TEXT NOT NULL,
    clinician_person_id TEXT NOT NULL,
    review_state TEXT NOT NULL CHECK (review_state IN ('reviewed', 'agreed', 'disagreed', 'needs_context')),
    note TEXT,
    created_at TEXT NOT NULL
);

```

6. Deterministic algorithm contract

for each authorized domain series:

1. preserve its native scale and improvement direction
2. resolve corrected/superseded evidence
3. require minimum observation count + time span
4. compute recent robust center (median) and variability
5. compare with prior window and prior direction
6. apply domain policy thresholds
7. require persistence for slowing/stalled/reversing
8. emit snapshot + evidence + limitations

Never:

- combine unrelated scales into one arithmetic score
- use missing data as zero
- label an engagement gap as predicted deterioration
- allow model prose to decide trajectory state

7. Personalized baseline

- The first version should compare the person to their own prior course, not to a population norm.
- A baseline/corridor is a descriptive reference window, not a promise of where the person "should" be.
- If future versions add normative/cohort expectations, they require a separate evidence/fairness validation path and must remain distinguishable from personal-history comparison.

8. Command Center integration

- Implement a recovery-trajectory AttentionSignalProvider using Handoff 03.
- Default signals: persistent reversing state; sustained stall on an active Return-to-Life goal; clinically meaningful slowing across multiple corroborating domains according to policy.
- Do not emit work for every stable domain. The Command Center is for action, not chart commentary.
- Signal language: "Recovery trajectory changed in sleep and function across the current review window" rather than "patient is off track" as an unexplained verdict.

9. Session Prep and patient view

- **Session Prep:** Changes and Trends section can say which domains moved, stalled, or reversed, with evidence and comparison window.
- **Clinician patient overview:** compact trajectory card with domain badges and a longitudinal chart link.

- **Trajectory detail:** native-scale lanes, goal/function lane, care/safety rails, and a written explanation of the state. Reconstructed data remains visually distinct.
- **Patient-facing:** do not expose "reversing" or "off track" labels without a dedicated patient-language design. Clinician v1 only.

10. AI role

Task	Purpose	Boundary
trajectory.explain	Turn deterministic snapshot into concise clinician wording	Cannot change state or thresholds.
trajectory.context_retrieve	Retrieve authorized concurrent threads/goals/interventions for context	Cannot state causation.
trajectory.session_prep_render	Compose multi-domain prep paragraph with citations	Conflicting domains remain explicit.

11. Events

Event	Meaning
trajectory.snapshot_computed.v1	Versioned domain state computed.
trajectory.deviation_proposed.v1	Actionable change qualifies for Command Center provider.
trajectory.reviewed.v1	Clinician records review/disagreement/context.
trajectory.deviation_resolved.v1	Previously actionable deviation no longer meets policy.

12. Implementation phases

Phase 1 - Domain adapters

Build

- Return-to-Life function series
- existing trajectory/check-in/measure adapters
- session response/recovery adapter

Definition of done

- No combined score
- native scales preserved

Phase 2 - Policy engine

Build

- Windowing
- robust comparison
- state machine
- limitations

Definition of done

- Single adverse point cannot create reversal under default policy
- insufficient data explicit

Phase 3 - UX + Session Prep

Build

- trajectory card/detail
- session prep adapter
- review recording

Definition of done

- Every state opens evidence

Phase 4 - Command Center provider

Build

- persistent deviation signal provider
- dedupe/lifecycle with Handoff 03

Definition of done

- No signal flood
- safety alerts remain separate

13. Acceptance criteria

- A clinician can see simultaneous improvement in one domain and worsening in another.
- Trajectory state is reproducible from evidence, cutoff, and policy version.
- A single check-in cannot silently become a recovery verdict.
- Return-to-Life goal history is a first-class function lane.
- Command Center receives only persistent/actionable deviations.
- No predictive risk or treatment-outcome claim is introduced.

Appendix A. File-level implementation map

Path	Change
src/lib/clinical/trajectory.ts	EXTEND adapters/metadata but preserve native-scale/provenance philosophy.
src/lib/clinical/recovery-trajectory.ts	NEW domain snapshot/state engine.
src/lib/clinical/trajectory-policy.ts	NEW versioned thresholds/window policy.
src/app/clinician/member/[id]/trajectory/page.tsx	NEW or extend existing record/progress surface.
src/components/clinical/RecoveryTrajectoryCard.tsx	NEW overview/session-prep card.
src/lib/clinical/attention-providers/recovery-trajectory.ts	NEW Handoff 03 provider.
src/lib/spine.ts + projections.ts	Add snapshot/review events as appropriate.
src/lib/db.ts + scripts/pg-schema.sql	Add/mirror schema.
tests/recovery-trajectory*.test.ts	Windowing, missingness, multi-domain conflict, security, replay, point-in-time.